Computer Networks Finals Cheat Sheet

Focus: Chapters 4-6 + Repeated Exam Concepts

Chapter 4: Network Layer - IP Addressing & Subnetting

IP Addressing Basics

- **32-bit address** divided into network and host parts
- **Classful Addressing**: Classes A (8-bit net), B (16-bit net), C (24-bit net), D (multicast), E (reserved)
- **CIDR/Classless Addressing**: Format A.B.C.D/n where n = prefix length
- **Slash notation**: 200.23.16.0/24 means first 24 bits are network

Subnetting Key Concepts

- **Subnet Mask**: 255.255.255.0 for /24, 255.255.255.192 for /26
- **Network address**: First address (host bits = 0)
- **Broadcast address**: Last address (host bits = 1)
- **Valid host addresses**: Network+1 to Broadcast-1
- **Number of addresses**: 2^{32-n} for /n notation
- **Borrowed bits** for subnetting: More 1s than default mask = subnet mask

Subnetting Example

- Organization gets 200.58.15.0/24 (256 addresses)
- Needs: 90, 33, 30, 17 addresses
  - Subnet 1: .0/25 (128 addresses, 90 hosts) → 200.58.15.0-127
  - Subnet 2: .128/26 (64 addresses, 33 hosts) → 200.58.15.128-191
  - Subnet 3: .192/27 (32 addresses, 30 hosts) → 200.58.15.192-223
  - Subnet 4: .224/28 (16 addresses, 17 hosts) → 200.58.15.224-239

Important Formulas

- Max hosts per subnet: $2^{host-bits} - 2$
- Number of subnets: $2^{borrowed-bits}$
- First usable IP: Network + 1
- Last usable IP: Broadcast - 1

Chapter 4: NAT & DHCP

NAT (Network Address Translation)

- **Purpose**: Allow multiple devices to share one public IP
- **Private address ranges**: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
- **Implementation**:
  - Replace source IP:port in outgoing packets
  - Maintain NAT translation table
  - Replace destination IP:port in incoming packets
- **Advantages**: Security (hides internal IPs), flexible ISP changes, device flexibility
- **Disadvantages**: Violates end-to-end principle, port manipulation by network layer

DHCP (Dynamic Host Configuration Protocol)

- **Steps**: Discover → Offer → Request → ACK
- **Server Discovery**: Broadcast to 255.255.255.255:67 with source 0.0.0.0:68
- **Provides**: IP address, subnet mask, default gateway, DNS server info
- **Leasing**: Addresses assigned for specific time period (3600 sec typical)
- **Key Ports**: Client 68, Server 67

Chapter 4: IPv6 & Tunneling

IPv6 Basics

- **Address Size**: 128 bits (vs 32 bits for IPv4)
- **Header**: Fixed 40-byte (vs variable for IPv4)
- **No checksum**: Faster processing at routers
- **No fragmentation**: Routers can't fragment (endpoints must handle)
- **Flow label**: Identify datagrams in same flow

Transition: IPv4 → IPv6

- **Tunneling**: IPv6 datagram carried as payload in IPv4 datagram
- **Dual-stack**: Routers run both IPv4 and IPv6
- **Encapsulation**: IPv6 packet wrapped in IPv4 packet between IPv4 networks

Chapter 4: Forwarding & Routing

Forwarding vs Routing

- **Forwarding**: Moving packets from input link to output link (data plane)
- **Routing**: Determining end-to-end paths (control plane)
- **Longest Prefix Matching**: Router selects most specific matching prefix in table

Router Architecture

- **Input ports**: Line termination, lookup, forwarding, queueing
- **Switching fabric**: Memory, bus, or interconnection network
- **Output ports**: Buffering, scheduling
- **Processing**: Nanosecond timeframe

Buffering Rule of Thumb

$$Buffer = RTT \times LinkCapacity = 250ms \times C$$

Chapter 5: Routing Protocols

OSPF (Open Shortest Path First)

- **Type**: Link-state, intra-AS routing
- **Algorithm**: Dijkstra's shortest path
- **Scope**: Within single AS
- **Metric**: Link cost (bandwidth, delay, etc.)
- **Convergence**: Faster than RIP
- **Advanced**: Hierarchical routing, authentication (MD5), multicast support
- **Uses**: Dijkstra - $O(n^2)$ complexity
- **Broadcasting**: Link-state advertisements flooded to entire AS

RIP (Routing Information Protocol)

- **Type**: Distance-vector, intra-AS routing
- **Metric**: Hop count (max 15)
- **Update interval**: 30 seconds
- **Convergence**: Slow, count-to-infinity problem
- **Advertisements**: Up to 25 subnets per message
- **Dead neighbor**: No advertisement for 180 sec
- **Poison reverse**: Prevent ping-pong loops

BGP (Border Gateway Protocol)

- **Type**: Exterior routing (inter-AS)
- **Variant**: eBGP (between ASes) and iBGP (within AS)
- **Approach**: Path-vector protocol
- **Connection**: Semi-permanent TCP connections (port 179)
- **Key Attributes**:
  - **AS-PATH**: ASes through which prefix has passed
  - **NEXT-HOP**: Internal AS router to reach next-hop AS
- **Route Selection** (priority order):
  1. Local preference (highest wins)
  2. Shortest AS-PATH
  3. Closest NEXT-HOP (hot potato routing)
  4. BGP identifier (tie-breaker)
- **Policy-based**: Based on business relations, not just performance

Dijkstra's Algorithm (Key Steps)

```
Initialize: D[u]=0, D[v]=∞ for all v
Loop:
  Find w not in N with min D[w]
  Add w to N
  For all neighbors v of w not in N:
    D[v] = min(D[v], D[w] + cost[w,v])
```

Bellman-Ford Equation (Distance Vector)

$$D_{xy} = \min_{v}(c_{x,v} + D_{vy})$$

- Iterative and distributed
- May have routing loops during convergence (count-to-infinity)
- Bad news travels slowly

Routing Protocols Comparison

Aspect	RIP	OSPF	BGP
Type	Distance-vector	Link-state	Path-vector
Scope	Intra-AS	Intra-AS	Inter-AS
Metric	Hop count	Link cost	Policy/AS-PATH
Convergence	Slow	Fast	Slow

Aspect	RIP	OSPF	BGP
Loops	Count-to-infinity	No loops	AS-PATH prevents loops
Max hops	15	Unlimited	Unlimited

Chapter 5: Virtual Circuits vs Datagrams

Virtual Circuit Network

- **Connection-oriented**: Setup, data transfer, teardown
- **VC numbers**: Change at each hop (from forwarding table)
- **Forwarding table**: Maps (incoming interface, VC#) → (outgoing interface, VC#)
- **Guaranteed service**: Predictable performance
- **Example**: ATM, Frame Relay

Datagram Network

- **Connectionless**: No setup required
- **Destination-based forwarding**: Based on destination IP only
- **Forwarding table**: Destination IP prefix → Output link
- **Best-effort**: No guarantees
- **Example**: Internet (IP)

VC Forwarding Table Example

```
Incoming Interface | Incoming VC | Outgoing Interface | Outgoing VC
1                  | 12          | 3                  | 22
2                  | 63          | 1                  | 18
```

Chapter 6: Link Layer & MAC Protocols

MAC Addresses

- **Format**: 48-bit (6 octets), hex notation (1A-2F-BB-76-09-AD)
- **Scope**: Locally unique within LAN only
- **Purpose**: Frame delivery within same subnet
- **Portability**: Can move between LANs, unlike IP

ARP (Address Resolution Protocol)

- **Function**: Map IP address to MAC address on same LAN
- **Operation**: Broadcast ARP query, unicast reply
- **TTL**: Typical 20 minutes for ARP table entries
- **Scope**: Only within same subnet

MAC Protocol Categories

1. Channel Partitioning

- **TDMA** (Time Division): Each station gets time slots
- **FDMA** (Frequency Division): Each station gets frequency bands
- **CDMA** (Code Division): Each station gets unique code
- **Efficiency**: Under-utilized at low traffic

2. Random Access

- **ALOHA**: Simple, collisions on overlap
- **Slotted ALOHA**: Better efficiency (37% max)
- **CSMA** (Carrier Sense Multiple Access): Listen before transmit
- **CSMA/CD**: Collision Detection (wired Ethernet)
  - Detect collision within time to detect
  - Send jam signal, back off
  - Binary exponential backoff: $K \in [0, 2^m - 1]$ after m collisions
  - Wait time: $K \times 512$ bit times
- **CSMA/CA**: Collision Avoidance (wireless 802.11)
  - RTS-CTS handshake for reservation
  - Defer if channel busy
  - Backoff after failed transmission

3. Taking Turns

- **Polling**: Central controller grants access
- **Token Passing**: Token circulates, node with token can transmit
- **Fairness**: Good at high load, overhead at low load

Ethernet (802.3)

- **CSMA/CD**: Collision detection
- **Standard frame format**: Preamble, Destination MAC, Source MAC, Type, Payload (46-1500 bytes), CRC
- **Speeds**: 10 Mbps (old) → 100 Mbps → 1 Gbps → 10 Gbps → 100 Gbps
- **Link types**: Copper (Cat5, Cat6), Fiber, Coax
- **Backward compatible**: Automatic rate negotiation

Ethernet CSMA/CD Algorithm

```
1. Sense channel - if idle, transmit full frame; if busy, wait
2. If entire frame sent without collision → Done
3. If collision detected → Abort, send jam signal
4. After m collisions, choose K ∈ [0, 2^m-1] randomly
5. Wait K×512 bit times, return to step 1
```

802.11 WiFi MAC (CSMA/CA)

- **No collision detection**: Hard in wireless
- **RTS-CTS exchange**: Sender reserves channel
  - RTS (Request to Send) → CTS (Clear to Send)
  - Prevents hidden terminal collisions
- **Backoff**: Random exponential backoff on failure
- **Frame format**: 4 MAC addresses (AP, sender, receiver, router)

Chapter 6: Ethernet Switches

Switch Basics

- **Function**: Connect LANs at link layer
- **Forwarding**: Based on MAC destination address only
- **Transparent**: Hosts unaware of switch presence
- **Plug-and-play**: No configuration needed
- **Full duplex**: Each link separate collision domain

Switch Self-Learning

- **Learning**: When frame received, learn sender's location (interface)
- **Flooding**: Unknown destination → forward to all ports except incoming
- **Filtering**: Known destination on same incoming segment → drop

- **Selective forwarding**: Forward only to necessary port
- **TTL-based expiration**: Entries expire after time period

Switching Table

```
MAC Address | Interface | TTL
A1:B2:C3    | 1         | 600
D4:E5:F6    | 3         | 600
```

Switches vs Routers

Feature	Switch	Router
Layer	Link (Layer 2)	Network (Layer 3)
Forwarding key	MAC address	IP address
Algorithm	Flooding/Learning	Routing protocols
Scope	Subnet	Across networks
Forwarding table	Switch table	Routing table

VLANs (Virtual LANs)

- **Port-based VLAN**: Ports grouped into virtual LANs
- **Efficiency**: Limits broadcast domain, improves security
- **Trunk ports**: Carry frames between VLAN-aware switches
- **802.1q protocol**: Adds VLAN header for trunk frames
- **Benefits**: Scaling, admin flexibility, security

Chapter 6: Error Detection & Correction

Parity Checking

- **Single-bit parity**: Detects single bit errors only
- **Two-dimensional parity**: Can detect and correct single bit errors
- **Method**: Checksum at end of data to detect flipped bits

CRC (Cyclic Redundancy Check)

- **Generator polynomial** G given
- **Calculation**: $D \cdot 2^r = Q \cdot G + R$ (all mod 2)
- **Transmission**: Send D followed by R
- **Reception**: Divide received frame by G, remainder should be 0

- **Error detection**: All burst errors less than r+1 bits detected
- **Example**: G=1001 (4 bits), r=3, D=101011
  - $D \cdot 2^3 = 101011000$
  - Divide by 1001 → R = remainder
  - Transmit: D || R

Internet Checksum

- **Method**: Sum 16-bit integers, take one's complement
- **Check**: Recompute at receiver, should equal received checksum
- **Limitation**: Detects errors but doesn't correct

Chapter 4-6: Important Exam Concepts

TCP Sequence Numbers & ACKs (Frequently tested)

- **Sequence number**: Number of first byte in segment
- **ACK number**: Next expected byte (cumulative)
- **Example**:
  - Receiver gets bytes 0-99 → sends ACK 100
  - Receiver gets bytes 100-199 → sends ACK 200

Routing Calculations (Very Important)

- **Link-state**: Dijkstra's algorithm, $O(n^2)$
- **Distance-vector**: Bellman-Ford, distributed
- **Shortest path**: Sum of link costs along path
- **Table format**: [Node | Distance | Via | Status]

Subnetting Calculations (Common Exam Questions)

- Given CIDR block, divide into required subnets
- Calculate: Network address, broadcast, first/last usable IPs
- Verify: No overlaps, sufficient addresses

Virtual Circuit Forwarding Table (Repeated question type)

- Path A→B: Specify VC numbers for each link
- Forward table shows: incoming (interface, VC) → outgoing (interface, VC)
- VC number can change at each hop

Forwarding vs Filtering (Switch operations)

- Forward to single port: Known destination
- Flood to all ports: Broadcast or unknown destination
- Filter (drop): Frame for device on same segment

BGP Route Selection (Multi-part question)

- Multiple routes to same destination
- Select based: local preference → AS-PATH length → NEXT-HOP cost → BGP ID
- Policy overrides performance

MAC Protocol Efficiency

- **Slotted ALOHA**: Max 37% at optimal p
- **CSMA**: Waits for idle, reduces collisions
- **CSMA/CD**: Also detects and aborts collisions
- **Polling/Token**: Fair, but overhead at low load

Error Detection Examples

- **CRC calculation**: Polynomial division with remainder
- **Parity**: Detect 1-bit (single), detect/correct 1-bit (two-dim)
- **Checksum**: Internet checksum for TCP/IP headers

Key Formulas & Constants

Concept	Formula
Addresses in subnet	2^{32-n} for /n notation
Host addresses	2^h - 2 (h = host bits)
Subnets created	2^b (b = borrowed bits)
Transmission delay	L/R (L=length bits, R=rate bps)
Throughput	$\min(R_c, R_s, R/N)$
Buffering	$RTT \times C$
Dijkstra time	$O(n^2)$
Slotted ALOHA efficiency	1/e ≈ 37%
CRC burst detection	All errors < r+1 bits

Quick Reference: Layer Operations

Network Layer (IP)

- Routing, forwarding, addressing, fragmentation, TTL
- Protocols: IP, ICMP, IGMP, OSPF, BGP

Link Layer (MAC)

- Frame delivery, error detection, MAC addressing
- Protocols: Ethernet, 802.11, ARP, VLANs

Key Performance Metrics

- **Latency**: Propagation + transmission + queueing + processing
- **Throughput**: Limited by bottleneck link
- **Fairness**: All flows get equal share at congestion
- **Loss**: Due to buffer overflow at routers

Study Tips for Finals

1. **Subnetting**: Practice dividing blocks, calculating addresses
2. **Routing**: Understand Dijkstra and Bellman-Ford differences
3. **MAC Protocols**: Know efficiency tradeoffs and collision handling
4. **Forwarding Tables**: Practice creating for routers, switches, VCs
5. **Calculations**: Time transmission, RTT, buffer requirements
6. **Protocol Comparison**: When to use each protocol type
7. **Error Detection**: CRC calculation, checksum verification
8. **Real examples**: DNS queries, ARP requests, TCP handshakes through routers [1] [2] [3]

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1. cn-pyqs_merged.pdf
2. cn_merged-notes_endsem.pdf
3. cn_merged-notes_endsem.pdf